

Camera Pose Estimation via NetVLAD

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Abstract

This work presents a visual place recognition and camera relocalization module that combines a pretrained feature extraction backbone with a differentiable NetVLAD aggregation layer. The framework learns discriminative global descriptors through a triplet margin loss and performs retrieval-based localization. We formalize the mathematical foundation, detail the implementation methodology, and conduct a detailed experimental analysis. Our codes can be found at https://github.com/ghafoorharis/2025_AAI5020_PA

1 Introduction

Global image descriptors enable efficient large-scale visual place recognition by encoding visual content into compact vector representations. The VLAD (Vector of Locally Aggregated Descriptors) framework aggregates local feature residuals, while NetVLAD provides a differentiable implementation suitable for end-to-end training on convolutional features. Our approach integrates frozen VGG16 features with a trainable NetVLAD head to construct a robust pipeline for camera relocalization through image retrieval and pose estimation. This work contributes a systematic implementation and analysis of the complete visual localization pipeline.

2 Methodology

2.1 Architecture Overview

We freeze VGG16 up to the last layer for dense feature extraction and stack a learnable NetVLAD layer that aggregates local descriptors into a unit-norm global representation. Our training uses a triplet margin loss to learn discriminative embedding and our inference utilizes cosine-similarity retrieval from a precomputed database. The final camera pose is obtained either by direct pose transfer from the top-ranked match or by geometric verification on the top- K candidates

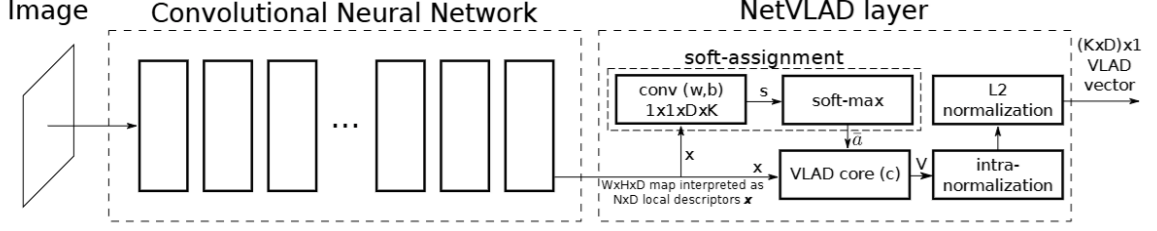


Figure 1: Network architecture: VGG16 extracts dense features that undergo descriptor-wise ℓ_2 normalization before NetVLAD aggregation. The NetVLAD layer performs soft assignment via a 1×1 convolution, residual summation, intra-normalization, and a final ℓ_2 normalization for retrieval.

2.2 Feature Extraction

Input images are resized to 224×224 and normalized using ImageNet statistics. The VGG16 backbone produces a tensor $X \in \mathbb{R}^{D \times H \times W}$ with $D=512$. Each spatial location yields a local descriptor $\mathbf{x}_i \in \mathbb{R}^D$ for $i = 1, \dots, N$ where $N=H \times W$. We apply descriptor-wise normalization $\mathbf{x}_i \leftarrow \mathbf{x}_i / \|\mathbf{x}_i\|_2$ to stabilize training.

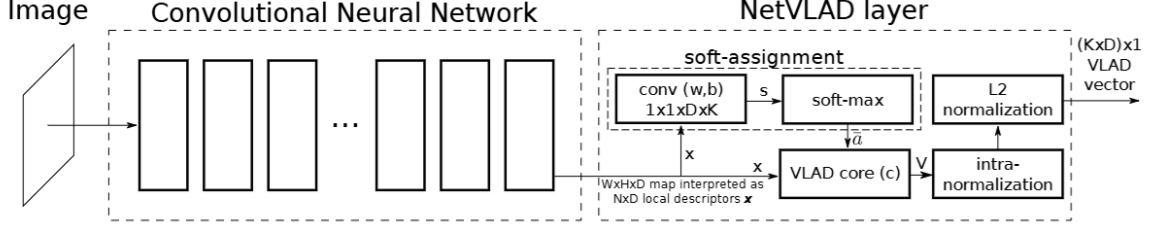


Figure 2: Network architecture: VGG16 extracts dense features that undergo descriptor-wise ℓ_2 normalization before NetVLAD aggregation. The NetVLAD layer performs soft assignment via a 1×1 convolution, residual summation, intra-normalization, and a final ℓ_2 normalization for retrieval.

2.3 NetVLAD Aggregation

As shown in figure 2, given the learnable cluster centers $\{\boldsymbol{\mu}_k\}_{k=1}^K$, a 1×1 convolution generates assignment logits $s_k(\mathbf{x}_i)$, which are normalized with a softmax:

$$a_k(\mathbf{x}_i) = \frac{\exp(s_k(\mathbf{x}_i))}{\sum_{j=1}^K \exp(s_j(\mathbf{x}_i))}. \quad (1)$$

Per-cluster residual vectors are

$$\mathbf{v}_k = \sum_{i=1}^N a_k(\mathbf{x}_i) (\mathbf{x}_i - \boldsymbol{\mu}_k) \in \mathbb{R}^D, \quad \mathbf{v} = [\mathbf{v}_1; \dots; \mathbf{v}_K] \in \mathbb{R}^{K \times D}. \quad (2)$$

Algorithm 1 NetVLAD Forward Pass

Inputs: Batch of images $x \in \mathbb{R}^{B \times 3 \times H_0 \times W_0}$; CNN feature extractor; cluster count K ; feature dim D ; learnable centroids $C \in \mathbb{R}^{K \times D}$; 1×1 conv producing assignment logits

Outputs: Global descriptors $V \in \mathbb{R}^{B \times (K \cdot D)}$

```
1:  $F \leftarrow \text{CNN\_FeatureExtractor}(x)$   $\triangleright F \in \mathbb{R}^{B \times D \times H \times W}$ 
2:  $N \leftarrow H \cdot W$ 
3:  $S \leftarrow \text{Conv}_{1 \times 1}(F)$  and reshape to  $S \in \mathbb{R}^{B \times K \times N}$ 
4:  $A \leftarrow \text{softmax}(S, \text{dim} = 1)$   $\triangleright$  soft-assignments per cluster,  $A \in \mathbb{R}^{B \times K \times N}$ 
5:  $X \leftarrow \text{reshape}(F, B \times D \times N)$ ;  $X^\top \leftarrow \text{transpose}(X, (0, 2, 1))$   $\triangleright X^\top \in \mathbb{R}^{B \times N \times D}$ 
6: Initialize  $V \leftarrow \mathbf{0}^{B \times K \times D}$ 
7: for  $k = 1$  to  $K$  do
8:    $c_k \leftarrow C[k] \in \mathbb{R}^D$ ; broadcast  $c_k$  to  $B \times N \times D$ 
9:    $a_k \leftarrow A[:, k, :] \in \mathbb{R}^{B \times N}$ ; reshape to  $B \times N \times 1$ 
10:   $R_k \leftarrow \sum_{n=1}^N a_k[:, n, 1] \cdot (X^\top[:, n, :] - c_k)$   $\triangleright$  residual aggregation
11:   $V[:, k, :] \leftarrow R_k$ 
12:  $V \leftarrow \text{normalize}(V, p = 2, \text{dim} = D)$   $\triangleright$  intra-normalization per cluster
13:  $V \leftarrow \text{reshape}(V, B \times K \cdot D)$ 
14:  $V \leftarrow \text{normalize}(V, p = 2, \text{dim} = 1)$   $\triangleright$  global L2 for retrieval
15: return  $V$ 
```

We apply *intra-normalization* ($\mathbf{v}_k \leftarrow \mathbf{v}_k / \|\mathbf{v}_k\|_2$) and a final global ℓ_2 normalization of $\mathbf{v}$. With unit-norm descriptors, cosine similarity is equivalent to the inner product and order-equivalent to squared Euclidean distance. We implement the NetVLAD layer as explained in algorithm 1.

2.4 Training Protocol

We train with the triplet margin loss. For triplets (q, p, n) and embeddings $(\mathbf{f}_q, \mathbf{f}_p, \mathbf{f}_n)$,

$$\mathcal{L} = \max\left(0, \|\mathbf{f}_q - \mathbf{f}_p\|_2^2 + m - \|\mathbf{f}_q - \mathbf{f}_n\|_2^2\right). \quad (3)$$

Because descriptors are unit-normalized, this enforces a margin in cosine similarity. We employ Adam (learning rate 10^{-4}), batch size 8, $K=16$ clusters, and a frozen VGG16 backbone.

2.5 Evaluation Methodology

We evaluate the image recognition by the metric Recall@K which is the fraction of queries for which at least one *positive* database image appears in the top- K . Positives are defined by a geographic distance threshold. The details of metric implementation is outlined in 2

Camera Relocalization. We evaluate two pose estimation strategies: (1) *Direct Transfer*: utilize the pose of the top-retrieved database image; (2) *Geometric Verification*: detect local features, establish correspondences, estimate essential matrix using RANSAC, recover relative

Algorithm 2 Recall@K for Image Retrieval

Inputs: Model $\mathcal{M}$; query dataset $\mathcal{D}_q$; database dataset $\mathcal{D}_b$; UTM coordinates $\{\text{utmQ}_i\}$, $\{\text{utmDb}_j\}$; distance threshold τ ; list K values

Outputs: Dictionary of recall scores $\text{Recalls}[k]$ for $k \in K$

- 1: $(E_q, I_q, P_q) \leftarrow \text{ComputeEmbeddings}(\mathcal{M}, \mathcal{D}_q)$
- 2: $(E_b, I_b, P_b) \leftarrow \text{ComputeEmbeddings}(\mathcal{M}, \mathcal{D}_b)$
- 3: $N_q \leftarrow |I_q|$, $N_b \leftarrow |I_b|$
- 4: Initialize $\text{Sim} \in \mathbb{R}^{N_q \times N_b}$, $\text{Dist} \in \mathbb{R}^{N_q \times N_b}$
- 5: **for** $i = 1$ to N_q **do**
- 6: **for** $j = 1$ to N_b **do**
- 7: $\text{Sim}[i, j] \leftarrow \text{cosine}(E_q[i], E_b[j])$
- 8: $\text{Dist}[i, j] \leftarrow \|\text{utmQ}_{I_q[i]} - \text{utmDb}_{I_b[j]}\|_2$
- 9: Initialize hits $H[k] \leftarrow 0$ for all $k \in K$
- 10: **for** $i = 1$ to N_q **do**
- 11: $\mathcal{P}_i \leftarrow \{j \mid \text{Dist}[i, j] < \tau\}$ $\triangleright$ ground-truth positives
- 12: **if** $|\mathcal{P}_i| = 0$ **then**
- 13: $\mathcal{P}_i \leftarrow \{\arg \min_j \text{Dist}[i, j]\}$ $\triangleright$ fallback
- 14: $\pi \leftarrow \text{argsort}(\text{Sim}[i, :])$ in descending order $\triangleright$ best to worst
- 15: **for** each $k \in K$ **do**
- 16: $\text{TopK} \leftarrow \pi[1:k]$
- 17: **if** $\mathcal{P}_i \cap \text{TopK} \neq \emptyset$ **then**
- 18: $H[k] \leftarrow H[k] + 1$
- 19: **for** each $k \in K$ **do**
- 20: $\text{Recalls}[k] \leftarrow 100 \cdot H[k] / N_q$
- 21: **return** Recalls

pose, and compose with database pose to obtain absolute camera orientation. Algorithm ?? details the geometric verification approach.

3 Experimental Analysis

3.1 Retrieval Performance

The Recall@K curve summarizes retrieval quality across K . At smaller K , performance reflects precision for the most relevant matches; larger K indicates whether a correct match appears among a broader candidate set, as shown in Fig. 3.

Figure 4 illustrates both the strengths and limitations of our NetVLAD descriptor approach. For queries with stable structural cues, multiple true positives appear in the top-5 despite viewpoint changes, confirming strong place-level invariance. Failure cases primarily occur in texture-poor scenes (sky-dominated views) and highly repetitive geometries where false positives exhibit cosine similarities comparable to true matches (≈ 0.98 – 0.99). The nearest

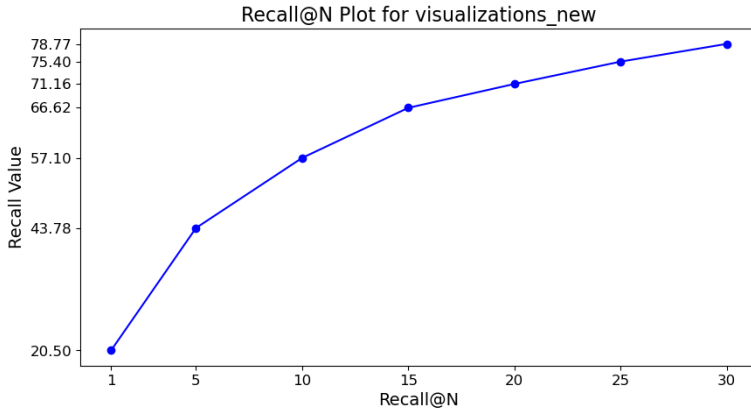


Figure 3: Retrieval performance: Recall@K on the test split.

ground truth often lies just outside the top-5, indicating small but consequential ranking errors. These observations motivate a lightweight geometric re-ranking stage on the top- K candidates to suppress near-duplicate confusions and reduce large-rotation outliers.

3.2 Camera Pose Estimation

We evaluate two distinct approaches for camera pose estimation as detailed in our methodology. Table 1 presents a comprehensive quantitative comparison of both approaches.

Table 1: Mean pose-estimation errors: direct transfer vs. geometric verification.

Metric	Direct Transfer	Geometric Verification
Translation Error (m)	12.23	10.48
Rotation Error (deg)	26.00	34.42

The geometric verification approach demonstrates improved translation estimation, reducing mean error by approximately 14% compared to direct pose transfer. However, rotation estimation shows increased error, suggesting challenges in robust essential matrix estimation from retrieved image pairs. This performance pattern indicates that while geometric constraints improve positional accuracy, rotation recovery remains sensitive to feature matching quality and viewpoint differences between query and database images.

4 Conclusion

This work presents a comprehensive implementation and analysis of a VGG16-NetVLAD architecture for visual place recognition and camera relocalization. The frozen backbone preserves general visual semantics while the trainable aggregation layer adapts to place-specific patterns. Future work should focus on optimizing the geometric verification pipeline



Figure 4: **Qualitative image-retrieval results.** Each row shows a query (left), the top-5 retrieved database images (middle), and the closest ground truth not appearing in the top-5 (right). Green frames denote true positives (TP) within the 25 m threshold; red frames denote false positives (FP); blue marks the nearest positive missed by the top-5. Numbers above each thumbnail report cosine similarity (Cos Sim) and ground-truth UTM distance (GT Dist, in meters).

through learned local features and adaptive RANSAC parameters, while exploring hybrid architectures that maintain the accuracy advantages demonstrated in our evaluation with reduced computational overhead.